Noah Peterson

515-393-7613 | petersonn249@gmail.com | linkedin.com/in/nwp249 | github.com/nwp249

SUMMARY

Computer Engineering student at Iowa State University with hands-on experience integrating software, electronics, sensors, and motor systems through robotics and embedded projects. Experienced with C/C++, Python, microcontrollers, PCB design, and hardware/software troubleshooting. Interested in embedded systems design and automated agricultural systems.

EDUCATION

Iowa State University, College of Engineering **GPA: 3.82**
B.S. Computer Engineering *Aug. 2024 – May 2027*
ChipForge ASIC Design Club — Member *Aug. 2026 – Present*

PROJECTS

Open-Source Car Head Unit | Senior Design | Rust/C++, Slint, Linux (SBC) **Sep. 2026 – May 2027**

- Defining system architecture for a standalone, open-source double-DIN car head unit built on an embedded Linux SBC with a touchscreen interface and required CAN bus integration.
- Scoping a modular, plugin-based software structure to support multiple audio inputs (Bluetooth, Wi-Fi, USB, analog), channeled audio output with digital signal processing (DSP), and community-driven extensibility.
- Establishing documentation and infrastructure-as-code standards so the software environment and hardware configuration is reproducible by outside contributors.

Remote Navigation Robot | ARM Cortex-M4 | C, Python **Nov. 2025 – Dec. 2025**

- Developed embedded firmware in C integrating sensors, motor controllers, and human interfaces for a mobile robotic system.
- Implemented robust error-handling to ensure system reliability and obstacle avoidance in real-time environments.
- Implemented UART and I2C communication protocols for data transmission between modules and system debugging.
- Designed a Python-based visualization tool to render 3D environmental data.

Trackball | KiCad | C **Jun. 2026 – Present**

- Designed a custom PCB to support wireless data translation via two ESP32-S3 microcontrollers, power management, and movement tracking.
- Reverse engineered a schematic for an optical sensor to support integration and data communication.

EXPERIENCE

Ecommerce Web Development **May 2026 – Present**
McClatchy Media *Des Moines, IA*

- Wrote automation software (JavaScript/Node.js) that manages three online stores from a single codebase and matches about 75,000 products to their source data with 95 to 97 percent accuracy.
- Reverse-engineered several undocumented web systems by analyzing their network traffic, then built reliable software around them. This included a background service that automatically renews login credentials expiring every 60 seconds, and scheduled jobs that send alerts when supplier or pricing data changes.
- Ran a full technical evaluation of a third-party data provider: tested over 1,000 links, estimated its database at roughly 40 million records, and found inaccurate stock data it was reporting as reliable.

Field Technician **Jun. 2022 – Aug. 2024**
Bayer Crop Science *Huxley, IA*

- Performed manual crop maintenance and weed control across research plots over three seasons, gaining direct, hands-on exposure to field trial operations and production agriculture.

TECHNICAL SKILLS

Languages: C, C++, Verilog, VHDL, Assembly (ARM, RISC-V), Python, Java

Embedded Systems: ARM Cortex-M4, Intel FPGA, UART, I2C

Hardware Design: KiCad (PCB layout, schematic capture), LTSpice

Software/Tools: Questa ModelSim, Quartus Prime, MS Word, Excel, PowerPoint